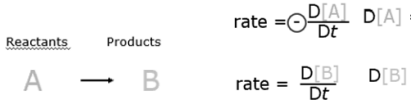
1. The Avogadro constant N_A is the ratio of the number of particles N of substance X to the amount of substance n of substance X. Its unit is mol^{-1}. It indicates how many particles are contained in one mole of particles. It is independent of the type of particle.	$$N_A = \frac{N(X)}{n(X)}$$ $$N_A = 6,022 \cdot 10^{23} \frac{1}{mol}$$ $$\Rightarrow n(X) = \frac{N(X)}{N_A}$$
2. The molar mass M of substance X is the ratio of the mass m of substance X to the amount of substance n of substance X. Its unit is g/mol.	$$M(X) = \frac{m(X)}{n(X)}$$ $$\Rightarrow n(X) = \frac{m(X)}{M(X)}$$
3. The molar volume V_m of an (ideal) gas is the ratio of the volume V of the gas particles X to the amount of substance n of gas particles X. Its unit is l/mol.	$$V_m = \frac{V(X)}{n(X)}$$ $$V_m = 22,41 \frac{l}{mol}$$ $$\Rightarrow n(X) = \frac{V(X)}{V_m}$$
4. The amount concentration (Molarity) C or M is the ratio of the amount of substance n of dissolved particles X to the volume of the solution $V_{solution}$. Its unit is mol/l.	$$Molarity\ (C) = \frac{n}{V} = \frac{moles\ solute}{L\ solution}$$ or $$Molarity\ (M) = \frac{n}{V} = \frac{moles\ solute}{L\ solution}$$
5. The average reaction rate v is the ratio of the change in concentration of the products Δc(Product) to the change in time Δt. It describes both the rate of formation of products and the rate of decomposition of reactants (with opposite sign).	
6. Law of Mass Action (LMA) The equilibrium constant Kc for reactions in solutions is the ratio of the product of concentrations c of products to that of reactants. The equilibrium constant KP for gas reactions is determined analogously from the partial pressures p of products and reactants. Solids are set equal to 1 in both cases. Coefficients enter as exponents.	$aA + bB \rightleftharpoons cC + dD$ $$Kc = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$
7. Gibbs-Helmholtz Equation The change in free enthalpy ΔG is the difference between the change in enthalpy ΔH and the product of the absolute temperature T (in K) and the change in entropy ΔS.	$\Delta G = \Delta H - T \cdot \Delta S$

8. **The ion product of water Kw**is the product of the concentrations of hydronium ions and hydroxide ions. Its unit is mol²/l². The pKW value is the negative base-10 logarithm of KW.	$$\begin{aligned} K_W &= c(H_3O^+) \cdot c(OH^-) = 10^{-14} mol^2/l^2 \\ pK_W &= -lg\{K_W\} = 14 \\ K_W &= K_S \cdot K_B \\ pK_W &= pK_S + pK_B \end{aligned}$$
9. Acid Constant K_S and pK_S The larger the acid constant KS, the stronger the acid. The pK_S value is the negative base-10 logarithm of K_S. The smaller the pK_S, the stronger the acid.	$$H_2O + HA \rightleftharpoons H_3O^+ + A^-$$ $$\begin{aligned} K_S &= \frac{c(H_3O^+) \cdot c(A^-)}{c(HA)} \\ pK_S &= -lg\{K_S\} \end{aligned}$$
10. Base Constant K_B and pK_B The larger the base constant KB, the stronger the base. The pKB value is the negative base-10 logarithm of KB. The smaller the pKB, the stronger the base.	$$B + H_2O \rightleftharpoons BH^+ + OH-$$ $$\begin{aligned} K_S &= \frac{c(BH^+) \cdot c(OH^-)}{c(B)} \\ pK_B &= -lg\{K_B\} \end{aligned}$$
11. The pH value is the negative base-10 logarithm of the hydronium ion concentration. The smaller the pH, the more acidic the solution.	pH = $-\log_{10}$ [H_3O^+] Or pH = $\log_{10}$ 1/[H_3O^+] . or pH = -log(c(H_3O^+))
12. **The pOH value** is the negative base-10 logarithm of the hydroxide ion concentration. The lower the pOH, the more basic the solution	pOH = -−$\log_{10}$ (c(OH^-)) ; pK_W = pH + pOH = 14
13. pH Value for Strong Acids For strong acids (complete dissociation), the approximation is:	$$c(H_3O^+) = c_0(HA) \quad ; \quad pH = -\log\left(c_0(HA)\right)$$
14. pH Value for Weak Acids For weak acids, the approximation is:	$$pH = \frac{1}{2} \cdot \left(pK_S - \log(c_0(HA))\right)$$
15. pH Value for Strong Bases For strong bases (complete dissociation), the approximation is:	$$pH = 14 + \log(c_0(B))$$
16. pH Value for Weak Bases For weak bases, the approximation is:	$$pH = 14 - \frac{1}{2} \cdot \left(pK_B - \log(c_0(B))\right)$$
17. Henderson-Hasselbalch Equation The pH value for buffer solutions of a weak acid HA and its conjugate base A^- is:	$$pH = pK_S + \log\left(\frac{c_0(A^-)}{c_0(HA)}\right)$$
18. Nernst Equation The redox potential E for a redox pair (half-cell/galvanic element: (Ox/Red) at 25°C is given by:	$$E = E^0 + \frac{0.0257}{n} \ln\left(\frac{[Ox]}{[Red]}\right)$$